



人造食用色素对健康有害，天然色素也不安全

Artificial food dyes aren't the only ones associated with increased risks of diabetes, cancer

人造食用色素并非唯一与糖尿病、癌症风险增加相关的色素。

Artificial food dyes have long been suspected to be harmful to your health. But new research shows that some of the natural color additives being turned to as alternatives are associated with an increased risk of Type 2 diabetes and cancer.

长期以来，人们一直怀疑人造食用色素对健康有害。但最新研究表明，作为替代品的一些天然色素添加剂与2型糖尿病和癌症风险增加有关。

Studies conducted by researchers in France showed that certain natural dyes were associated with a more than 40% increased risk of Type 2 diabetes and certain cancers, in research published this week. The scientists, from several French universities and research institutions, looked at what more than 100,000 people ate and drank, including specific brands, and followed participants for up to eight years on average.

本周发表的研究显示，法国研究人员进行的研究表明，某些天然色素与患2型糖尿病和某些癌症的风险增加40%以上有关。这些来自法国多所大学和研究机构的科学家调查了超过10万人的饮食情况，包括特定品牌，并对参与者进行了平均长达八年的追踪。

The Trump administration is pushing America's food giants to embrace natural food colors that it views as safer alternatives to artificial dyes. It is a core goal of Health Secretary Robert F. Kennedy Jr. and his Make America Healthy Again movement. Big food companies have in turn been swapping in natural dyes, citing shifting consumer preferences.

特朗普(Trump)政府正推动美国食品巨头采用天然食用色素，认为它们是比人造色素安全的替代品。这是卫生部长小罗伯特·肯尼迪(Robert F. Kennedy Jr.)及其“让美国再次健康”(MAHA)运动的核心目标。大型食品公司也以消费者偏好转变为由，纷纷改用天然色素。

The push away from artificial colors in favor of natural ones has spurred a flurry of activity in the past year or so. The Food and Drug Administration has approved the use of several new natural food colors since last May. Some states, including California, West Virginia and Utah, have passed laws restricting the sale of foods containing certain artificial dyes. Many more states are considering similar legislation. Big food companies including General Mills and WK Kellogg, which makes Froot Loops cereal, have announced plans to remove some synthetic dyes from their products.

过去一年左右，弃用人造色素转而使用天然色素的趋势引发了一系列行动。自去年5月以来，美国食品药品监督管理局(FDA)已批准使用几种新的天然食用色素。包括加利福尼亚州、西弗吉尼亚州和犹他州在内的一些州已通过法律，限制销售含有某些人造色素的食品。还有更多州正考虑类似的立法。包括通用磨坊(General Mills)和生产Froot Loops麦片的WK Kellogg在内的大型食品公司已宣布计划从产品中去除部分合成色素。

But the studies, conducted by researchers from several French universities and research institutions, showed that both synthetic and natural colors are linked to health problems.

但由法国多所大学和研究机构的研究人员进行的这些研究表明，合成色素和天然色素都与健康问题有关。

“Natural food additives are not a proof of safety for consumers,” said Mathilde Touvier, research director at the National Institute of Health and Medical Research in France and the studies’ lead author.

“天然食品添加剂并不能保证对消费者安全,” 法国国家健康与医学研究院(National Institute of Health and Medical Research)的研究主任、上述研究的主要作者玛蒂尔德·图维耶(Mathilde Touvier)说。

Several natural color additives were associated with increased likelihood of developing Type 2 diabetes, including beta-carotene, which can be extracted from carrots or red palm oil and is added as a color to foods like cheese, yogurt, packaged cakes and fruit drinks. People who had higher intakes of beta-carotene as a food coloring had a 44% increased risk of Type 2 diabetes, compared with those who had a lower intake.

几种天然色素添加剂与患2型糖尿病的可能性增加有关, 其中包括β-胡萝卜素。这种色素可从胡萝卜或红棕榈油中提取, 并作为色素添加到奶酪、酸奶、包装蛋糕和果汁饮料等食品中。与摄入量较低的人相比, 将β-胡萝卜素作为食用色素摄入量较高的人患2型糖尿病的风险增加了44%。

Among the other natural color additives associated with a higher risk of Type 2 diabetes were curcumin (49% increased risk) and anthocyanins (40% increased risk), which can be processed from fruits and vegetables. The study was published this week in Diabetes Care, a journal from the American Diabetes Association.

其他与患2型糖尿病风险增加相关的天然色素添加剂包括姜黄素和花青素, 两者带来的风险分别增加49%和40%, 这些色素可以从水果和蔬菜中加工提取。该研究本周发表在美国糖尿病协会(American Diabetes Association)的期刊Diabetes Care上。

In another study, food color additives, synthetic and natural, were associated with a 21% increased risk of breast cancer. The natural color additive beta-carotene was linked to a 41% increased risk of breast cancer.

另一项研究发现, 合成和天然的食用色素添加剂与患乳腺癌风险增加21%有关。天然色素添加剂β-胡萝卜素与患乳腺癌风险增加41%有关。

The natural color dubbed ordinary or plain caramel was associated with a 15% increased risk of overall cancer. This kind of caramel is often used in baked goods and ice cream. The study was published in the European Journal of Epidemiology.

被称为普通焦糖的天然色素与整体癌症风险增加15%有关。这种焦糖常用于烘焙食品和冰淇淋中。该研究发表在《欧洲流行病学杂志》(European Journal of Epidemiology)上。

Touvier said it may have been surprising that the natural food color additives yielded similar risks as synthetic ones. But several experiments have shown that substances may behave very differently in the body when stripped from the original food and processed into an additive. Some emerging animal and laboratory research has found that several color additives can spur inflammation and may cause insulin resistance. The substances may also cause dysfunction in the gut microbiome.

图维耶表示，天然食用色素添加剂会带来与合成色素相似的风险，这或许令人惊讶。但多项实验表明，当某些物质从原始食物中被剥离并加工成添加剂时，它们在体内的表现可能会大不相同。一些最新的动物和实验室研究发现，几种色素添加剂会诱发炎症，并可能引发胰岛素抵抗，这些物质还可能造成肠道菌群失调。

The studies are part of a larger project called the NutriNet-Santé cohort that is examining diet's effects on health. To home in on the impact of the specific additives, the researchers adjusted the results for many factors that could affect health including the nutritional quality of the food and the presence of other additives.

这些研究是一个名为 NutriNet-Santé 队列的更大项目的一部分，该项目旨在研究饮食对健康的影响。为了集中研究特定添加剂的影响，研究人员针对可能影响健康的许多因素对结果进行了调整，这些因素包括食品的营养质量以及是否含有其他添加剂。

Such observational research alone doesn't prove that natural food colors harm health. Touvier said the message for consumers is to eat fewer ultraprocessed foods. Food color additives aren't necessary, she said.

仅凭这种观察性研究并不能证明天然食用色素会损害健康。图维耶说，给消费者的建议是少吃超加工食品。她表示，食用色素添加剂并不是必需的。

“These are purely cosmetic, so it’s just here to make the product more attractive, more beautiful, and allow manufacturers to make us want to consume more,” she said.

“这些纯粹是为了美观，所以它们只是为了让产品更具吸引力、更好看，并让制造商能够激发我们增加消费的欲望，”她说。

Natural Food Colors Embraced by MAHA Linked to Health Problems

By Andrea Petersen